

Why Computational Social Sciences?

Asking Big Questions with Code

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Computational Social Sciences

Way of study human behavior like the natural sciences: **measure** at scale, **simulate** it, **experiment**, and **test** our ideas.

That is genuinely new. Large scale use of data has been confined to subfields inside SS fields (e.g., political methodology).

But it is still social sciences: if you want, **CSS is social sciences on steroids**.

We did all that before, but now, we can do at a scale that was not previously thought possible.

This talk: five ways CSS changed the way we do social sciences.

Computational Social Science (CSS)

Three promises

1. Computational Social Science (CSS) is **learnable** you already have part of what you need.
2. It is **useful in real research**, not just a buzzword.
3. The field is **young** **the good questions are still wide open.**

I will show you **five areas** of CSS, with real studies and live code.
All areas that you will learn in this Summer Institute.

Example of things we can do with modern computing

In 2010, Facebook showed **61 million** users a single “I Voted” button on election day.

A randomized experiment, embedded in a product, run on a population larger than most countries.

. . . it measurably increased **real-world turnout** — and the effect spread to **close friends** who never saw the message (small per person, large in aggregate).

No survey. No lab. A social-science experiment at planetary scale. (Bond et al., Nature 2012)

That is the kind of question CSS lets we answer.

So what *is* Computational Social Science?

A working definition

The use of **computation** (large data, algorithms, and simulation) to ask and answer questions about **human and social behavior**.

It lives at the intersection of three traditions:

Social science — the *questions*, the *theory*, the *meaning*.

Computer science — the *scale*, the *algorithms*, the *engineering*.

Statistics — the *inference*: separating signal from noise, cause from coincidence.

Why now? Three forces converged

Data

Society became digital. Every click, post, payment, and trip leaves a trace.

Compute

A laptop today out-muscles a 1990s supercomputer. The cloud is rentable by the hour.

Methods

Machine learning, NLP, and now LLMs turn messy human stuff into measurable quantities.

None of these is new alone. Their **collision** — roughly the last 15 years — is what made CSS a field.

What CSS actually does

The recurring pattern

Start with a real question about people or society.

Measure it at scale, or simulate it, or predict it.

Doubt the result until it earns your trust.

The computation is learnable.

The judgment — knowing when a measurement is fooling you — is the part that takes real work.

The five engines of CSS

We will tour five ways computation powers social inquiry:

1. **Text as data**: reading millions of documents at once.
2. **Networks**: relationships as the unit of analysis.
3. **Digital traces**: the data exhaust of online life.
4. **Simulation & causal ML**: growing societies, and finding causes.
5. **LLMs & AI**: language models as instruments and as subjects.

For each: *what it is, a real study, a peek under the hood, and why it matters.*

And this here is a tiny and incomplete version of what Drs Alam, Bilal, Khan, Qazi, Pasta, Samad, and Usmani will do.

Every document is data

Humanity's record is mostly **text**: laws, speeches, news, contracts, reviews, tweets, court rulings, diplomatic cables.

For centuries, reading it didn't scale: one scholar, one career, a few thousand pages.

The shift

Turn words into numbers, and a question that took a lifetime takes an afternoon across **millions** of documents, consistently.

This is **text-as-data**: counting, classifying, and measuring meaning at scale.

Showcase: measuring the latent from the visible

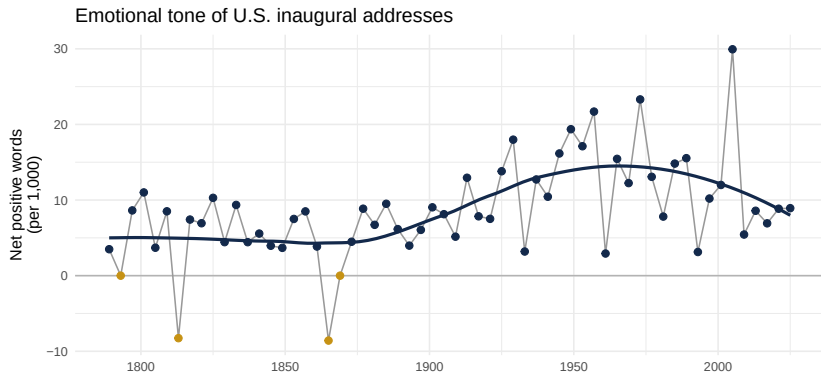
We never observe “tone,” “ideology,” or “polarization” directly. Text lets us **estimate** them.

Gentzkow, Shapiro & Taddy (2019): measured the **partisanship of speech** in the U.S. Congress and showed it exploded after the 1990s.

Media slant, central-bank “hawkishness,” judicial sentiment, propaganda detection, all now routinely measured from text.

Below: a tiny version you can run today — the emotional **tone** of every U.S. inaugural address, 1789-2021.

Live: tone of two centuries of inaugural addresses



Real corpus (quanteda) but a toy 25-word dictionary — an illustration of the *method*, not a validated finding.

Under the hood (it's shorter than you think)

```
library(quanteda)

# 1. Documents in, tokens out
toks <- tokens(my_corpus, remove_punct = TRUE)

# 2. A dictionary of concepts you care about
dict <- dictionary(list(
  fear = c("threat", "danger", "crisis"),
  hope = c("freedom", "progress", "unity")))

# 3. Count concepts per document -> a number you can model
scores <- dfm(tokens_lookup(toks, dict))
```

Dictionaries are the *training wheels*. The frontier: **topic models**, **word embeddings**, and **transformers** that learn meaning from context.

Why text as data matters

What changes

You can read the [whole archive](#), not a hand-coded sample.

You can [measure](#) things we used to only argue about tone, framing, ideology.

It is [reproducible](#): the codebook is code, and anyone can rerun it.

The catch: A number from text is still a claim. It has to be validated.

Relationships are the data

Classic statistics assumes your subjects are **independent**. Society is the opposite: we are defined by who we know, trade with, cite, follow, and fight.

The network lens

Stop studying **nodes** in isolation. Study the **ties** and the structure they form.

The questions change: not “who is influential?” but “*where* does influence flow?” Not “is this group cohesive?” but “*how* is it held together?”

Showcase: structure you can't see from inside

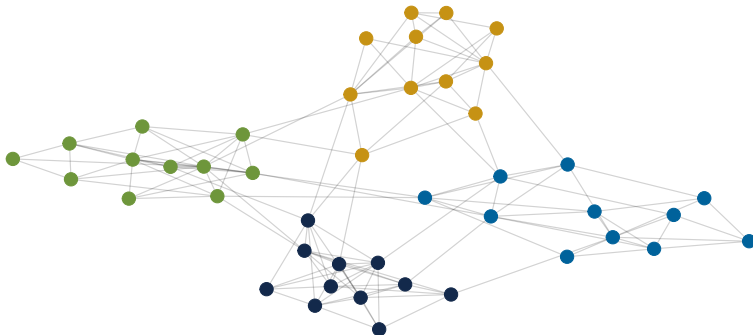
Diffusion: a famous study claimed obesity and smoking “spread” through friendship ties — even to friends-of-friends (Christakis & Fowler). *We come back to this.*

Politics: legislative co-sponsorship and donation networks reveal coalitions no roll-call vote shows directly.

States & trade: the world economy as a network predicts which sanctions bite and which crises cascade.

The recurring magic: **community detection** finds groups nobody labeled — emergent structure, discovered from ties alone.

Live: finding hidden communities



No group labels were given. The algorithm (`cluster_louvain`) recovered four communities from the wiring alone.

Under the hood

```
library(igraph)

# A graph from a list of ties (who -> whom)
g <- graph_from_data_frame(edges)

# Who sits at the crossroads of information flow?
btw <- betweenness(g)

# What groups emerge from the structure itself?
groups <- cluster_louvain(g)
```

Three lines take you from a spreadsheet of relationships to **centrality** (who matters) and **communities** (how the system is organized).

Why networks matter

What changes

You stop assuming people are **independent** and model how they actually depend on one another.

You can **discover** coalitions and brokers nobody labeled.

You can **trace** how behavior, money, and information spread.

The catch: spread through a network and a shared environment look alike.
Telling them apart is hard.

But be careful: A famous finding that fell apart

The headline

"Obesity is **contagious**. It spreads through your friends, even friends-of-friends."

Christakis & Fowler, *NEJM* 2007

Massive data, massive press, yet the result **does not hold up**:

Homophily (we befriend people like us) and **contagion** (friends change us) are *generically confounded* in observational networks — Shalizi & Thomas (2011).

Correlation across ties cannot separate **influence** from **selection**.

The algorithm found the pattern; only **causal reasoning** could tell it wasn't real.
That is why social science matters. (See Cunningham, *Causal Inference: The Mixtape*.)

The data exhaust of modern life

Every day we generate **found data** — created for other purposes, but priceless for research:

Search queries, web traffic, Wikipedia edits.

Mobile-phone location, transit taps, satellite imagery.

Prices scraped from a million online stores.

Digned vs. found

Surveys are **designed** clean, but small and expensive.

Traces are **found** gigantic and cheap, but messy and biased.

CSS is largely the craft of turning found data into trustworthy answers.

Showcase: measuring the previously unmeasurable

Satellites as economists: nighttime lights track GDP growth where official statistics are absent or doctored (Henderson, Storeygard & Weil 2012).

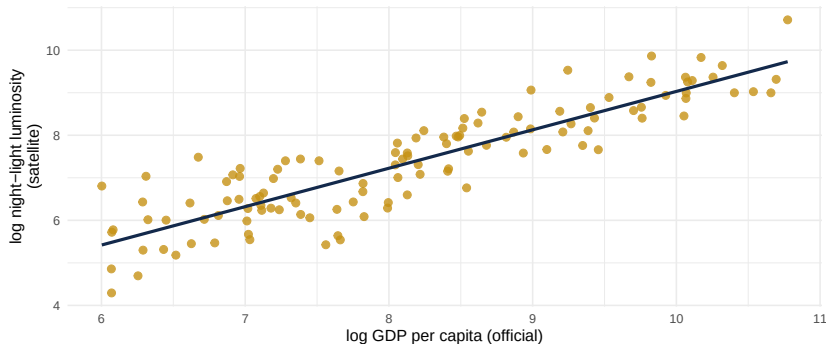
The Billion Prices Project: scraped online prices measured inflation in real time — and caught a government under-reporting it (Argentina).

The Opportunity Atlas (Chetty et al.): de-identified tax records mapped economic mobility down to the *neighborhood*.

. . . and a cautionary tale: **Google Flu Trends** predicted flu from searches — then drifted badly. *Big data is not self-correcting.*

Live: satellites as a measuring instrument

When official numbers are missing, light can stand in



Illustrative (simulated points). The real lights–GDP elasticity is *estimated* from data — not the slope drawn here.

Why digital traces matter

What changes

You can reach **places and people** surveys never could.

You get **real-time**, fine-grained measurement not a yearly snapshot.

You can bring **new evidence** to old, stuck debates.

The hard part is never getting the data. It's earning the right to believe it.

When you can't run the world twice

Two of social science's deepest problems:

We can rarely run the **experiment** (you can't randomize who gets a revolution).

We can rarely see the **counterfactual** (what would have happened otherwise?).

Two computational answers

Simulation build an artificial society and watch it run.

Causal ML squeeze causal estimates from observational data, and find *who* an effect falls on.

Showcase: Schelling's segregation model

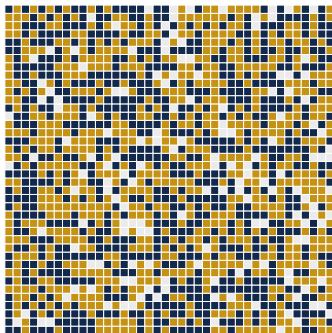
Thomas Schelling asked in 1971: can a city become **sharply segregated** even if no one wants it to be?

His agents are mildly tolerant: each is content as long as **at least a third** of its neighbors are like itself. Nobody seeks segregation.

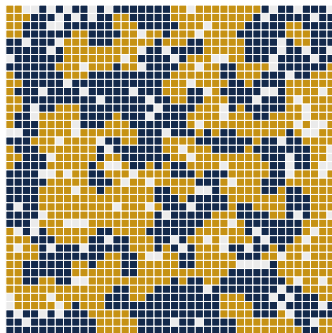
Watch what happens anyway.

Live: mild preferences, stark outcomes

Start: well mixed



End: self-sorted



empty A B

Nobody wanted segregation. Everybody got it. **Emergence**: macro patterns no individual intended.

The other half: machine learning for *causes*

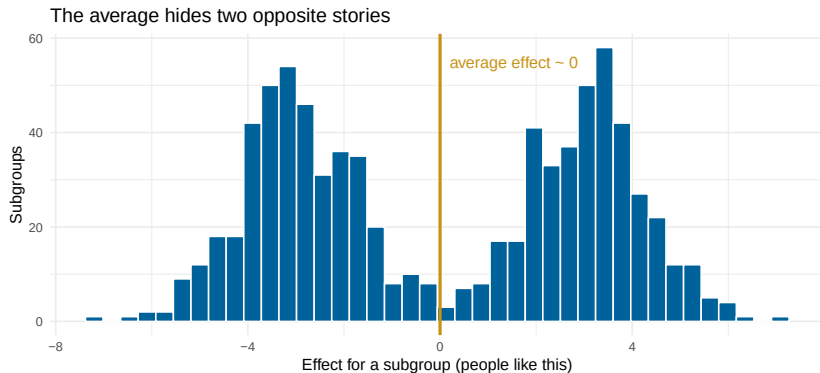
Prediction asks *what*. Social science usually asks *why* and *for whom*.

Double / debiased ML uses flexible models to strip out confounding, then estimates a clean causal effect.

Causal forests estimate **heterogeneous** effects — not “does the program work?” but “*whom does it work for?*”

A policy with **zero** average effect can still be transformative for a subgroup — and ML can find them.

Live: a zero average can hide everything



Some subgroups helped, others harmed; the average is near zero. Causal ML estimates these *subgroup* effects — not any one person's.

Why simulation & causal ML matter

What changes

You can **test theories** you could never run as a real experiment.
You move from "does it work **on average**?" to "**for whom** does it work?"

You make **emergence** and feedback explicit instead of hand-waving them.

The catch: a simulation is only as honest as its assumptions.

Language models as scientific instruments

The newest engine. Large language models are useful to social science in **two** very different ways.

1. As a measuring instrument

Point an LLM at a million documents and ask it to read, classify, and code them work that once needed an army of research assistants.

2. As a simulated subject

Ask an LLM to *role-play* a survey respondent or an agent a cheap, fast, controversial new kind of "data."

Showcase: the promise and the asterisk

LLMs now **annotate text** — stance, emotion, topic — at near-human accuracy for a fraction of the cost (with careful validation).

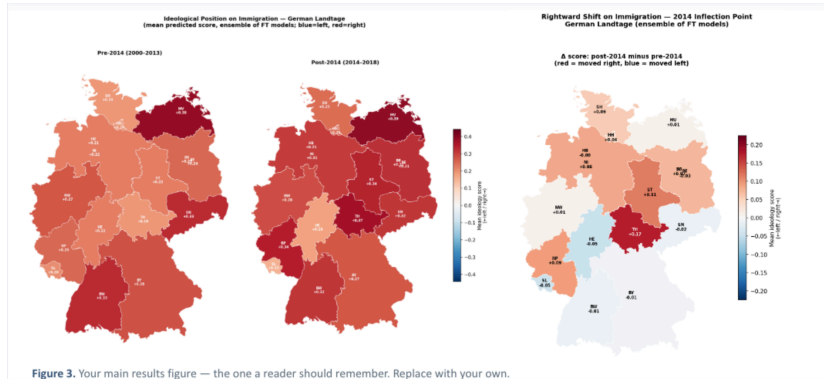
“**Silicon samples**”: prompting an LLM with a demographic profile reproduces some patterns of human survey responses (Argyle et al.) — useful for piloting, dangerous if trusted blindly.

Generative agent societies: populations of LLM agents that remember, plan, and interact — a new laboratory for social dynamics (Park et al.).

The non-negotiable rule

An LLM measurement is a **hypothesis**, not a fact. *Always validate against human-coded ground truth.*

A working example: ideal points from text



Open-source LLMs place German state legislatures (*Landtage*) on a left–right immigration scale — and reveal a [rightward shift after 2014](#). Mignozzetti, Xiao, Xu & Yu (2026), “*Ideal-Point Estimation Using Open-Source Large Language Models*” (working paper).

The dangers, stated plainly

LLMs are powerful and **treacherous** as research tools:

Hallucination — confident, fluent, wrong.

Bias — they encode the web's prejudices; “silicon samples” mirror who is over-represented online.

Opacity & drift — closed models change under you; results stop replicating.

Contamination — the model may have *read the answer key* during training.

The skill is no longer *getting* an answer. It is knowing **when to believe it** — exactly the social scientist's craft, now mission-critical.

Why LLMs matter — and where they bite

What changes

You can **measure subtle concepts** stance, emotion, framing at scale.

You can **pilot** a study in hours instead of months.

But the model is a **mirror of the web**: confident, biased, sometimes wrong.

Non-negotiable: validate against real, human-coded ground truth. Always.

First, the honest part: pitfalls

Computation makes you powerful. It also makes mistakes **faster and bigger**.

Big data good data. A million biased points is still biased — just confidently so.

Found data has hidden holes. Who is *missing* from the trace?

p-hacking at scale. Test enough hypotheses and noise will reward you.

Privacy & ethics. Just because data is *available* doesn't make it *yours* to use.

Reproducibility. If it isn't scripted and shared, it didn't happen.

Credibility is the whole game. The methods are only as good as your skepticism.

A starter toolkit

Languages

Python text, ML, scraping,
LLMs

R stats, causal inference, viz

Workhorse libraries

Text: quanteda, spaCy,
HuggingFace

Networks: igraph,

NetworkX

ML/causal: scikit-learn,
grf, EconML

Learn the craft

Bit by Bit, Salganik (free
online)

Text as Data, Grimmer,
Roberts & Stewart

Summer schools: SICSS

Data to play with

Wikipedia, GDELT,

Twitter/X archives

Census & World Bank open
data

All that you are going to learn in this Summer Institute.

Recap: the five engines

1. **Text as data** read the whole archive; measure the latent.
2. **Networks** relationships are the data; structure emerges.
3. **Digital traces** found data, at the cost of vigilance.
4. **Simulation & causal ML** grow societies; find who an effect falls on.
5. **LLMs & AI** powerful instruments, only as good as your validation.

Five different engines, one underlying move: make a social question measurable, then test it honestly.

Thank you — now go try it

Computational Social Science is not a club you have to qualify for.

It is a **habit**: bring a real question, pick up one tool, and start small.

Email me a question you'd like to answer
I'll point you at the data and the method.

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